

Applied Data Science Project with Python

Project Guidelines — Academic Year 2026–2027

Project philosophy and purpose

The baseline concepts and techniques covered in class serve merely as your starting point. The core objective of this project is to challenge you to **go beyond foundational learning** exposed in the Python module lessons by cultivating an autonomous engineering and research mindset. You are expected to actively explore new development methods, investigate and select the most suitable analytical or modeling approaches for your dataset, and independently drive technical solutions. Generative AI tools are welcome to support this discovery process, provided you maintain a rigorous, critical mindset regarding every line of code and technical decision.

1 General Regulations and Submission Guidelines

- **Group Organization:** The project must be completed in teams of **3-4 students**.
- **Topic Selection:** Each group must select a distinct project proposal from the 20 offered datasets. Topics are assigned on a first-come, first-served basis. **Can we select a different topic away from these 20 proposals by ourselves?** Absolutely **YES**. **BUT**, please, reach me via email (victor.navas@uab.cat) before start working to check the viability and suitability of your project proposal.
- **Submission Format:** Deliver access to a private repository (GitHub or GitLab) with instructor permissions granted. The repository must contain:
 1. **Source Code ('/src' or Jupyter Notebooks):** Fully reproducible, well-commented, and executable without dependency errors. Emphasize clean structure, modularity, and efficient vectorized operations with `pandas` and `numpy`.
 2. **Executive Report / Manuscript (PDF):** A structured, well-written document covering data handling justifications, exploratory data analysis (EDA), visualizations, modelling and critical discussion of results.
 3. **Environment Specification:** A valid `requirements.txt`, `pyproject.toml`, or `environment.yml` file.

2 Required Report Structure

To ensure the project is self-contained and reflects a real-world scientific workflow, the final report must explicitly include the following sections:

1. **Introduction & Problem Statement:** Contextual overview of the data, problem categorization (e.g., supervised classification, regression, clustering), analytical hypotheses, and team goals.
2. **Data Cleaning & Feature Engineering (Data Handling):** Detailed justification of strategies applied for missing value imputation, outlier handling, categorical encoding, feature scaling, and data transformations.
3. **Exploratory Data Analysis & Visual Storytelling:** High-quality graphics built with `seaborn`, `matplotlib`, or `plotly`, interpreting domain-specific insights, distributions, and feature relationships.
4. **Theoretical Foundations & Machine Learning Justification:** Explicit theoretical explanation of the chosen Machine Learning algorithms. You must justify why these specific models are methodologically suited for the underlying data distribution, task constraints, and problem dynamics.
5. **Model Implementation & Performance Evaluation:** Rigorous experimentation comparing baseline models against advanced architectures. Discussion of hyperparameter tuning, validation strategy, and selection of domain-appropriate evaluation metrics.
6. **Challenges, Enablers, & Lessons Learned:** Critical reflection detailing technical or analytical obstacles encountered during execution, model trade-offs made, and key tools that facilitated development.
7. **Individual Contribution Breakdown:** A clear table or statement outlining the specific role, responsibilities, and tasks completed by each of the group members.
8. **Transparent Generative AI Declaration:** Mandatory log of AI usage and critical evaluation of AI outputs.

3 Transparent policy on Artificial Intelligence (AI) usage

The use of Generative AI tools (such as ChatGPT, GitHub Copilot, Claude, etc.) as development and learning assistants is permitted. However, to maintain academic integrity and promote critical thinking, the following **mandatory policy** applies:

Requirements for AI Usage

1. **Full Transparency:** Include a dedicated appendix titled “*Declaration of Generative AI Tool Usage*”.
2. **Log of Prompts and Responses:** Document which tools were used, key prompts entered, and the specific scope of assistance (e.g., code refactoring, bug fixing, generating visualization ideas).
3. **Critical Validation:** The team remains fully responsible for code accuracy, efficiency, and prevention of AI hallucinations. In the appendix, provide at least one example where AI-generated code or proposal was incorrect or suboptimal, detailing how your team identified and corrected the issue.

4 Project Proposals (Kaggle Datasets)

Each team will select one topic. Proposals combine deep data handling with Machine Learning modeling and theoretical justification:

1. Telecom Customer Churn: Data Pipeline & Attrition Modeling

Task Type: Supervised Learning – Binary Classification

Dataset: [Kaggle - Telco Customer Churn](#)

Statement: Construct a clean, reproducible Python data pipeline to transform customer usage data and conduct deep exploratory analysis to identify attrition drivers. Furthermore, build an end-to-end classification framework to predict customer churn, exposing the theoretical foundations of chosen algorithms (e.g., Logistic Regression vs. XGBoost) and evaluating performance under class imbalance.

2. Ames Housing: Advanced Data Cleaning & Property Valuation

Task Type: Supervised Learning – Multivariate Regression

Dataset: [Kaggle - Ames Housing Dataset](#)

Statement: Focus on advanced missing value imputation, continuous feature scaling, and categorical encoding using efficient Pandas logic. Develop a regression framework to estimate property prices, theoretically justifying the choice between regularized linear models (Lasso/Ridge) and non-linear ensemble methods.

3. Credit Card Fraud: Robust Wrangling & Anomaly Detection

Task Type: Supervised Learning – Imbalanced Classification / Anomaly Detection

Dataset: [Kaggle - Credit Card Fraud Detection](#)

Statement: Implement custom Python functions for feature scaling, outlier filtering, and data profiling on highly imbalanced transactions. Address class imbalance theoretically and empirically using cost-sensitive classifiers or anomaly detection algorithms (e.g., Isolation Forests), prioritizing Precision-Recall metrics.

4. NYC Airbnb: Spatial Data Handling & Rental Price Modeling

Task Type: Supervised Learning – Regression & Spatial Feature Mining

Dataset: [Kaggle - NYC Airbnb Open Data](#)

Statement: Clean, parse, and aggregate short-term rental listings across neighborhood groups using optimized Pandas code. Model listing prices by engineering spatial and text-derived features, theoretically contrasting parametric linear models with tree-based algorithms.

5. E-Commerce Reviews: NLP Preprocessing & Text Classification

Task Type: Natural Language Processing (NLP) – Text Classification

Dataset: [Kaggle - Women's E-Commerce Clothing Reviews](#)

Statement: Build a modular text-processing pipeline (cleaning, tokenization, stop-word removal, TF-IDF/embeddings) from scratch. Evaluate and justify NLP classification architectures (e.g., Naive Bayes vs. Gradient Boosting) to predict review ratings or recommendation statuses.

6. Rossmann Store Sales: Time-Series Data Handling & Forecasting

Task Type: Time-Series Forecasting / Supervised Regression

Dataset: [Kaggle - Rossmann Store Sales](#)

Statement: Master complex date/time operations, lag feature creation, and multi-table merging in Pandas. Build predictive sales models, exposing the theoretical considerations of time-series cross-validation and temporal feature engineering.

7. IBM HR Analytics: Data Quality Audit & Attrition Modeling

Task Type: Supervised Learning – Binary Classification & Explainable AI

Dataset: [Kaggle - IBM HR Analytics Employee Attrition](#)

Statement: Conduct a comprehensive data quality audit and demographic feature analysis. Train and tune classification models, providing a strong theoretical justification for model selection and leveraging explainability tools (e.g., SHAP, feature importance) to analyze turnover risk.

8. Diabetes Health Indicators: Medical Handling & Risk Prediction

Task Type: Supervised Learning – Classification & Disease Risk Assessment

Dataset: [Kaggle - Diabetes Health Indicators](#)

Statement: Handle non-trivial medical distributions, categorical encoding, and automated profile generation. Formulate a classification pipeline to predict diabetes risk, theoretically contrasting parametric vs. non-parametric models with focus on clinical sensitivity/recall.

9. Spotify Audio Features: Dimensionality Reduction & Song Clustering

Task Type: Unsupervised Learning – Clustering & Dimensionality Reduction

Dataset: [Kaggle - Spotify Tracks Dataset](#)

Statement: Implement a modular workflow to normalize audio metrics and handle high-dimensional spaces with NumPy/Pandas. Apply dimensionality reduction (PCA, t-SNE) and unsupervised clustering (K-Means, DBSCAN), grounding algorithm selection in mathematical and structural theory.

10. US Air Quality: Multi-Station Aggregation & Pollution Forecasting

Task Type: Supervised Regression / Spatiotemporal Modeling

Dataset: [Kaggle - US Pollution Data](#)

Statement: Merge and clean heterogeneous air quality measurements using window functions and rolling averages. Develop spatiotemporal regression models to forecast air quality indices, detailing the theoretical assumptions behind spatial/temporal features.

11. Used Cars Market: Text Parsing & Automated Price Estimation

Task Type: Supervised Learning – Regression with Unstructured Parsing

Dataset: [Kaggle - Craigslist Used Cars Dataset](#)

Statement: Extract and clean noisy text and numeric data (unstructured specs, mileage strings). Build custom string parsing utilities, handle severe missingness, and implement non-linear regression models to capture car price depreciation curves.

12. Online Retail II: Transaction Processing & Customer Segmentation

Task Type: Unsupervised Clustering & Value Prediction

Dataset: [Kaggle - Online Retail II Dataset](#)

Statement: Process high-volume transaction logs and write scalable Pandas routines for RFM (Recency, Frequency, Monetary) aggregations. Segment buyer personas using clustering algorithms and build predictive models for customer value, theoretically framing customer lifetime modeling.

13. Netflix Catalog: Content Analytics & Categorical Pattern Discovery

Task Type: Unsupervised Learning / Pattern Mining & Classification

Dataset: [Kaggle - Netflix Movies and TV Shows](#)

Statement: Parse nested lists (cast, genres, countries) and clean temporal fields. Implement categorical vectorization, apply clustering or pattern mining, and theoretically justify models that group or classify unlabelled catalog content.

14. World Happiness: Multi-Source Integration & Socioeconomic Regression

Task Type: Supervised Learning – Multivariate Regression

Dataset: [Kaggle - World Happiness Report](#)

Statement: Combine multiple yearly reports, handle changing survey schemas, and perform robust data alignment. Model national happiness scores, theoretically addressing multicollinearity and evaluating linear vs. non-linear regression techniques.

15. **US Commercial Flight Delays: Large-Scale Handling & Delay Prediction**
Task Type: Supervised Learning – Large-Scale Classification / Regression
Dataset: [Kaggle - Flight Delays and Cancellations](#)
Statement: Handle large-scale tabular datasets efficiently using Pandas memory optimization techniques. Process temporal features, extract delay metrics, and evaluate predictive classifiers to forecast flight delay status with theoretical justification.
16. **Video Game Sales: Data Standardization & Global Trend Prediction**
Task Type: Supervised Learning – Multi-Target / Regional Regression
Dataset: [Kaggle - Video Game Sales](#)
Statement: Standardize sales metrics across regional markets, handle missing publisher information, and perform temporal aggregations. Build predictive models for regional/global success, comparing parametric vs. tree-based regression approaches.
17. **Heart Failure Clinical Records: Survival Profiling & Clinical Classification**
Task Type: Supervised Learning – Binary Classification / Clinical Risk
Dataset: [Kaggle - Heart Failure Clinical Records](#)
Statement: Conduct statistical profiling, check for collinearity, and create feature transformation pipelines. Train mortality classification models, providing theoretical backing for clinical metric optimization (Sensitivity vs. Specificity).
18. **Global Superstore Sales: Retail Analytics & Profitability Forecasting**
Task Type: Time-Series / Supervised Regression
Dataset: [Kaggle - Superstore Sales Dataset](#)
Statement: Build an automated data pipeline to parse order dates, calculate profit margins, and aggregate discount impacts. Construct predictive regression/forecasting models, theoretically analyzing discount interaction terms and seasonal trends.
19. **Stack Overflow Developer Survey: Complex Processing & Salary Modeling**
Task Type: Supervised Learning – Regression & Salary Profiling
Dataset: [Kaggle - Stack Overflow Annual Developer Survey](#)
Statement: Handle complex multi-option survey responses and semi-structured categorical variables. Build reusable parsing functions, encode technical stacks, and evaluate regression models to predict developer compensation from noisy survey data.
20. **Amazon Products & Pricing: E-Commerce Audit & Rating Prediction**
Task Type: Supervised Learning – Multi-Output Regression / Classification
Dataset: [Kaggle - Amazon Products Dataset](#)
Statement: Audit messy e-commerce listings containing dirty price strings and missing descriptions. Write regular expressions to extract numeric fields, structure catalog data, and build ML models predicting product ratings or price tiers.

5 Assessment Rubric

The project grade accounts for a maximum score of 10 points according to the following evaluation rubric:

Criteria	Weight	Description and Standards
Data Engineering & Handling	25%	Exceptional management of missing values, outliers, data types, scaling, and feature engineering. Efficient, vectorized, and modular Python code.
Exploratory Analysis & Storytelling	20%	Clear, elegant, and insightful visualizations. Extraction of actionable, non-obvious domain insights and target correlations.
ML Foundations & Model Justification	25%	Explicit theoretical justification for algorithm choices, thorough model evaluation, baseline comparison, and metric selection aligned with problem constraints.
Scientific Rigor & AI Usage	15%	Autonomous exploration beyond classroom basics, critical evaluation of GenAI tool outputs, and documented technical problem-solving.
Reproducibility & Organization	15%	Well-structured repository, valid environment specification, clean code architecture, and transparent individual contribution breakdown.

6 Date

You must submit your projects (software and report) no later than 23:59 on **December 9th, 2026**.

7 Contact

Office Hours & Contact Information

If you have any questions regarding dataset selection, project scope, or submission requirements, feel free to reach out via email:

- Víctor Navas-Portella
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